



In silico protocols for reproducibility of machine learning approaches

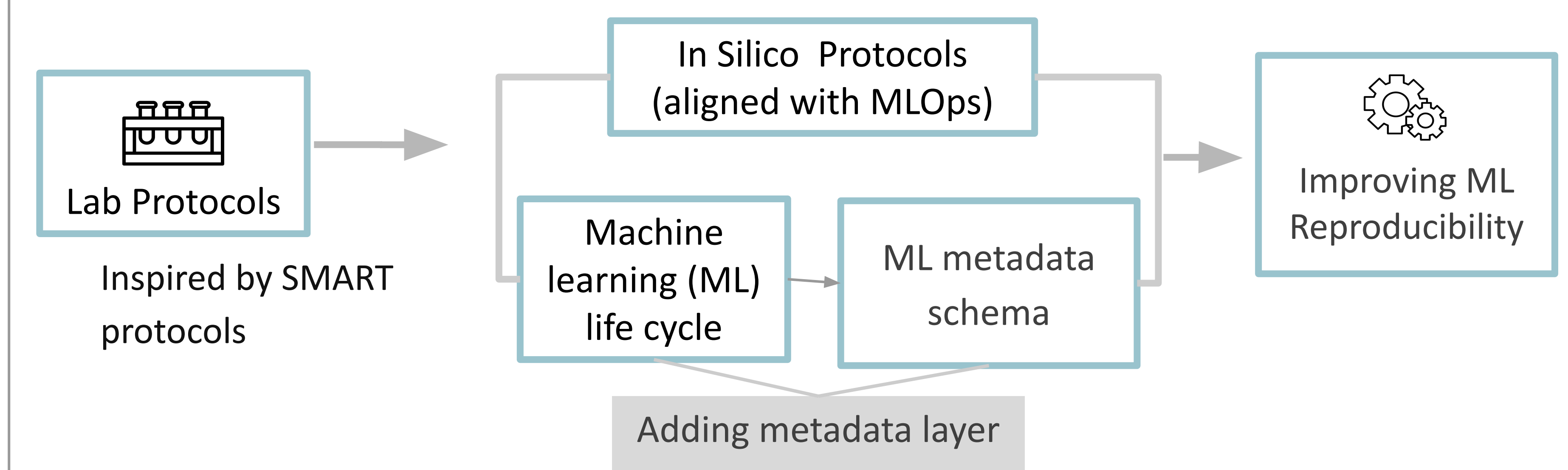
Dhwani Solanki¹, Leyla Jael Castro¹ and Dietrich Rebholz-Schuhmann^{1,2}

¹ ZB MED Information Centre for Life Sciences, Cologne, Germany, ² University of Cologne, Cologne, Germany

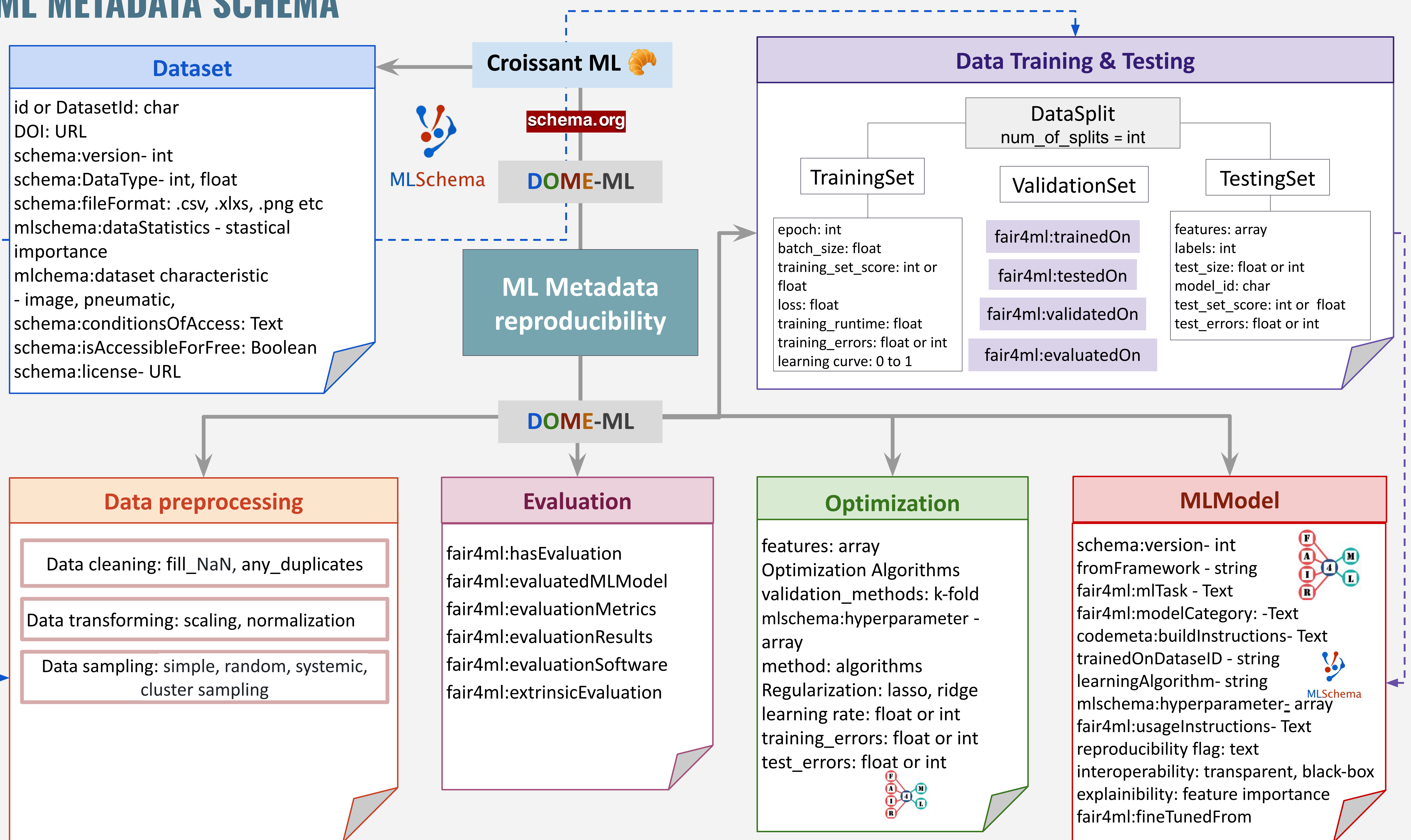
INTRODUCTION

- Lab protocols: systematic representation of tasks and operations
- Inspired by lab protocols and the metadata layer proposed by SMART protocols.
- It increases reproducibility & reusability
- Introducing metadata layer to Machine Learning (ML Experiments)
- in-silico protocol is aligned to the ML cycle commonly considered in MLOp
- From pre-processing to Model evaluation
- Integration of various metadata efforts

METHODS

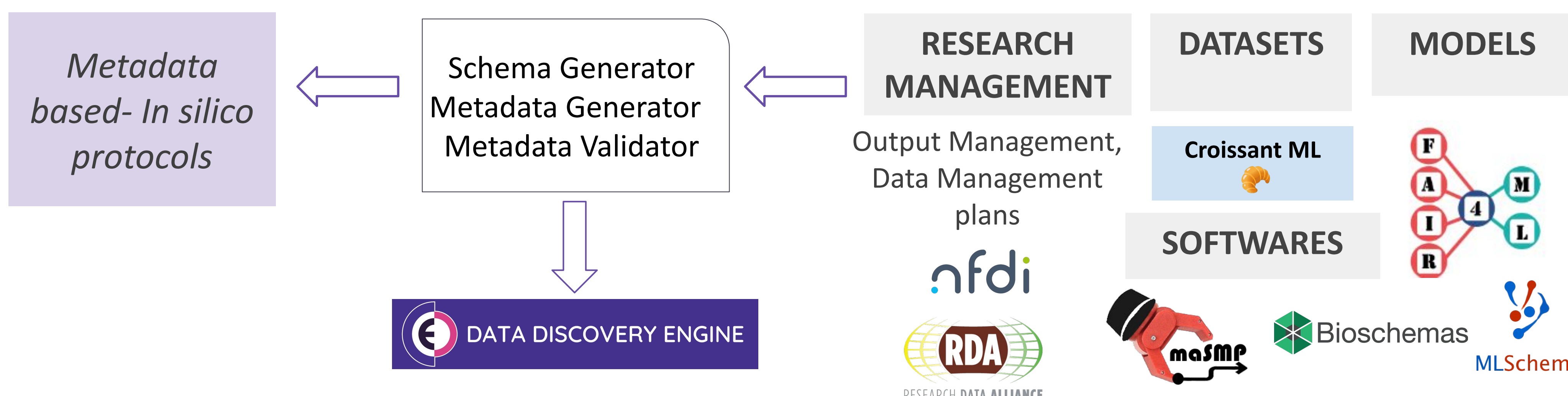


ML METADATA SCHEMA



FUTURE ACTIONS

Bringing together necessary vocabularies as a form of an ontology to facilitate reproducibility through DDE platform



REFERENCES

1. Giraldo, O., García, A., López, F. *et al.* Using semantics for representing experimental protocols. *J Biomed Semant* 8, 52 (2017). <https://doi.org/10.1186/s13326-017-0160-y>
2. Giraldo, O., García A., and Corcho, O.. 2014. SMART protocols: semantic representation for experimental protocols. In Proceedings of the 4th International Conference on Linked Science - Volume 1282. CEUR-WS.org, Aachen, DE, 36–47. https://ceur-ws.org/Vol-1282/lisc2014_submission_2.pdf
3. <https://rda-fair4ml.github.io/FAIR4ML-schema/release/0.1.0/index.html>
4. <https://zbmed-semtec.github.io/maSMPs/>
5. <https://mlcommons.org/working-groups/data/croissant/>
6. <https://github.com/ML-Schema/core>

